SepSys

Transforming Unstructured Sepsis Literature into Structured Validated Evidence

By Token_burners

SepSys

Clinical sepsis research is buried in PDFs. SepSys extracts it – producing structured, source-grounded, analysis-ready evidence tables from full-text publications.

Structured Evidence

Consistent, schema-validated fields

Analysis-Ready

Tables ready for meta-analysis

Source-Grounded

Every claim traceable to its origin

Verifiable

Auditable extraction outputs

Processing steps

- PDF Ingestion
- Table extraction
- Text extraction
- Image extraction
- Data aggregation
- Chunking
- Chunks processing
- Final output

Layout-Aware PDF Ingestion

Full clinical PDFs are parsed using **Docling**, preserving the scientific document structure that naive text extraction would destroy.

Sections

Introduction, Methods, Results

Tables

Data grids and result matrices

Figures

Captions and visual metadata

Page Metadata

Page numbers and layout context

Structure-Aware Chunking

Naive chunking breaks clinical context. SepSys uses section-aware and sentence-aware chunking strategies to keep related evidence together.

Outcomes & Effect Sizes

Odds ratios, hazard ratios, and AUC values stay co-located with their parent result statement.

Tables

Tables are usually missed by naïve chunking but Docling preserves format

Cohort Descriptions

Patient population details remain bound to the study context they describe.

Traceability & Verification

Every chunk carries a full provenance record – enabling exact source grounding and auditable extraction workflows.

Paper ID + Section

Each chunk is linked to its originating publication and document section.

Page & Chunk Index

Precise location metadata enables direct navigation back to the source.

Evidence Verification

Claims can be independently verified against the original manuscript at any time.

End-to-End Pipeline

Six stages transform raw clinical PDFs into verified, analysis-ready evidence – with full provenance preserved at every step.

INPUT: PDFs
(materials/articl
es/*.pd)

→

STAGE 1 — OCR + PARSING
(Docling + PyMuPDF)
Text + headings + page numbers
Tables → Markdown/JSON
Figures → cropped images
Output:
parsed_papers/{paper_id}.json

→

STAGE 2 — EXPORT
Sections → .md
Tables → JSON
Output: .md + _tables.json

→

STAGE 3 — CHUNKING
Structure-aware splitting (heading / paragraph / sentence)
Metadata: page, section, chunk_id
Output: _chunks.json

→

STAGE 4 — EMBEDDING + VECTOR DB (Weaviate + MiniLM)
Chunk embeddings (L2-normalized vectors)
Stored in Weaviate collection

→

STAGE 5 — RETRIEVAL (RAG)
BM25 keyword search
Vector similarity search
Hybrid fusion + optional BGE reranker
Output: Top-K relevant chunks

→

STAGE 6 — FRONTEND
(Streamlit + Dashboard)
Ingestion UI (parse / chunk / ingest)
Chat interface (RAG queries)
Analytics + traceability view

Retrieval Flow (Chat Tab)

```
User query
     │
     ▼
Embedder.embed(query)  →  float[384]
     │
     ├──── BM25 ───────────────────────────────┐
     │     Weaviate keyword search             │
     │                                         │
     ├──── Vector ─────────────────────────────┤  top-N candidates
     │     cosine similarity                   │
     │                                         │
     └──── Hybrid (default) ───────────────────┘
           Weaviate score fusion
                │
                ▼
          [Optional] Reranker
          CrossEncoder(query, chunk_text) per candidate
          → sorted by cross-encoder score
                │
                ▼
          top-K chunks returned
```

Analysis-Ready Output

Extractions are materialized as structured evidence tables – each row is a fully cited, field-validated finding ready for meta-analysis or clinical review.

Study	Population	Sample Size	Predictor	Outcome	Timing	Method	Effect Size	Performance	Notes	Source Anchor	Page
Current study	Critically ill children aged 1 month to 15 years admitted to PICU >24 hours	N=286	p-SOFA score	30-day mortality	Within 24 hours of admission	ROC curve analysis; sensitivity, specificity, accuracy	AUC=0.81 (95% CI 0.76–0.86), p=0.001	Sensitivity 93.87%, Specificity 38.21%, Accuracy 69.93% (cut-off >2)	p-SOFA better accuracy than PRISM III 24; scores within 24h	Maximum p-SOFA AUC=0.81 (0.76–0.86)	1
El-Mashad et al.	PICU children	Not reported	p-SOFA score	30-day mortality	Not reported	ROC curve analysis	AUC=0.89	Sensitivity 100%, Specificity 5.2% (cut-off >2); Sensitivity 80.9%, Specificity 81.8% (cut-off >6.5)	SOFA more accurate than PRISM; multiple cut-offs evaluated	SOFA >2 predicted death in 40% PICU patients	6
Matics et al.	PICU infants	Not reported	SOFA score	30-day mortality	Not reported	ROC curve analysis	AUC=0.88	Not reported	Applied SOFA in infant PICU population	SOFA AUC 0.88 for 30-day mortality prediction	6
Lalitha et al.	PICU children	Not reported	p-SOFA score	Mortality	Not reported	ROC curve analysis	AUC=0.84	Not reported	Compared p-SOFA vs PRISM III	p-SOFA AUC 0.84 vs PRISM III 0.70	6

.json and .md chunks contain paper metadata from the ingestion level, enabling LLM outputs to be traced back.

BM25 vs Vector search

BM25: **term-frequency based ranking function.**

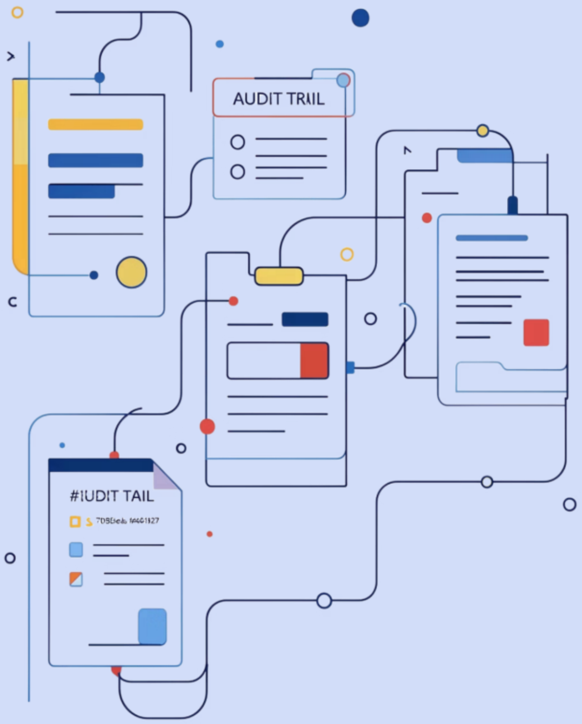
Benefit: Operates on exact token level. Good for retrieving exact variable names and values.

Vector/Cosine similarity:

Embeddings in a multi-dim space. Match on semantic similarity

Benefit: Good for semantic or paraphrasing questions or for connecting higher-level concepts.

Weaviate allows us to switch one way or the other !



Frontend

Frontend working on ingested and chunked outputs. Gives useful, effective, insightful visualizations.
Post-processing .

Attribute Analysis, Knowledge Graphs, Cross-study consensus	Traceability & Explainability

Literature Atlas

Interactive PDF viewer with section to description and networkx json support.